AIM:Simulation of TDMA-based MAC Protocol
Simulate TDMA slot allocation and compare energy efficiency with CSMA.
CODE:

```
# ======================================================
# TDMA Style MAC Simulation with NAM Animation
# NS2.35
# ======================================================

set val(chan)       Channel/WirelessChannel
set val(prop)       Propagation/TwoRayGround
set val(netif)      Phy/WirelessPhy
set val(mac)        Mac/802_11
set val(ifq)        Queue/DropTail/PriQueue
set val(ll)         LL
set val(ant)        Antenna/OmniAntenna
set val(ifqlen)     50
set val(nn)         4
set val(rp)         DSDV
set val(x)          500
set val(y)          500

set ns [new Simulator]

# Trace file
set tracefd [open tdma.tr w]
$ns trace-all $tracefd

# NAM file
set namfd [open tdma.nam w]
$ns namtrace-all-wireless $namfd $val(x) $val(y)

# Topography
set topo [new Topography]
$topo load_flatgrid $val(x) $val(y)

create-god $val(nn)

# Wireless Channel
set chan_1_ [new $val(chan)]

# Node Configuration
$ns node-config \
   -adhocRouting $val(rp) \
   -llType $val(ll) \
   -macType $val(mac) \
   -ifqType $val(ifq) \
   -ifqLen $val(ifqlen) \
   -antType $val(ant) \
   -propType $val(prop) \
```

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```
    -phyType $val(netif) \
    -channel $chan_1_ \
    -topoInstance $topo \
    -agentTrace ON \
    -routerTrace ON \
    -macTrace ON

# Create Nodes
for {set i 0} {$i < $val(nn)} {incr i} {
    set node_(i)[ns node]
}

# Node Positions
$node_(0) set X_ 50
$node_(0) set Y_ 250
$node_(0) set Z_ 0

$node_(1) set X_ 150
$node_(1) set Y_ 250
$node_(1) set Z_ 0

$node_(2) set X_ 250
$node_(2) set Y_ 250
$node_(2) set Z_ 0

$node_(3) set X_ 350
$node_(3) set Y_ 250
$node_(3) set Z_ 0

for {set i 0} {$i < $val(nn)} {incr i} {
    $ns initial_node_pos node(i) 20
}

# ================================================
# FLOW 1 : Node0 -> Node3
# ================================================

set udp0 [new Agent/UDP]
$ns attach-agent $node_(0) $udp0

set null0 [new Agent/Null]
$ns attach-agent $node_(3) $null0

$ns connect $udp0 $null0

set cbr0 [new Application/Traffic/CBR]
$cbr0 set packetSize_ 512
$cbr0 set interval_ 0.05
$cbr0 attach-agent $udp0
```

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```
# ================================================
# FLOW 2 : Node1 -> Node3
# ================================================

set udp1 [new Agent/UDP]
$ns attach-agent $node_(1) $udp1

set null1 [new Agent/Null]
$ns attach-agent $node_(3) $null1

$ns connect $udp1 $null1

set cbr1 [new Application/Traffic/CBR]
$cbr1 set packetSize_ 512
$cbr1 set interval_ 0.05
$cbr1 attach-agent $udp1

# ================================================
# FLOW 3 : Node2 -> Node3
# ================================================

set udp2 [new Agent/UDP]
$ns attach-agent $node_(2) $udp2

set null2 [new Agent/Null]
$ns attach-agent $node_(3) $null2

$ns connect $udp2 $null2

set cbr2 [new Application/Traffic/CBR]
$cbr2 set packetSize_ 512
$cbr2 set interval_ 0.05
$cbr2 attach-agent $udp2

# ================================================
# TDMA SLOT ALLOCATION
# ================================================

# Slot 1 : Node0 transmits
nsat1.0"cbr0 start"
nsat1.5"cbr0 stop"

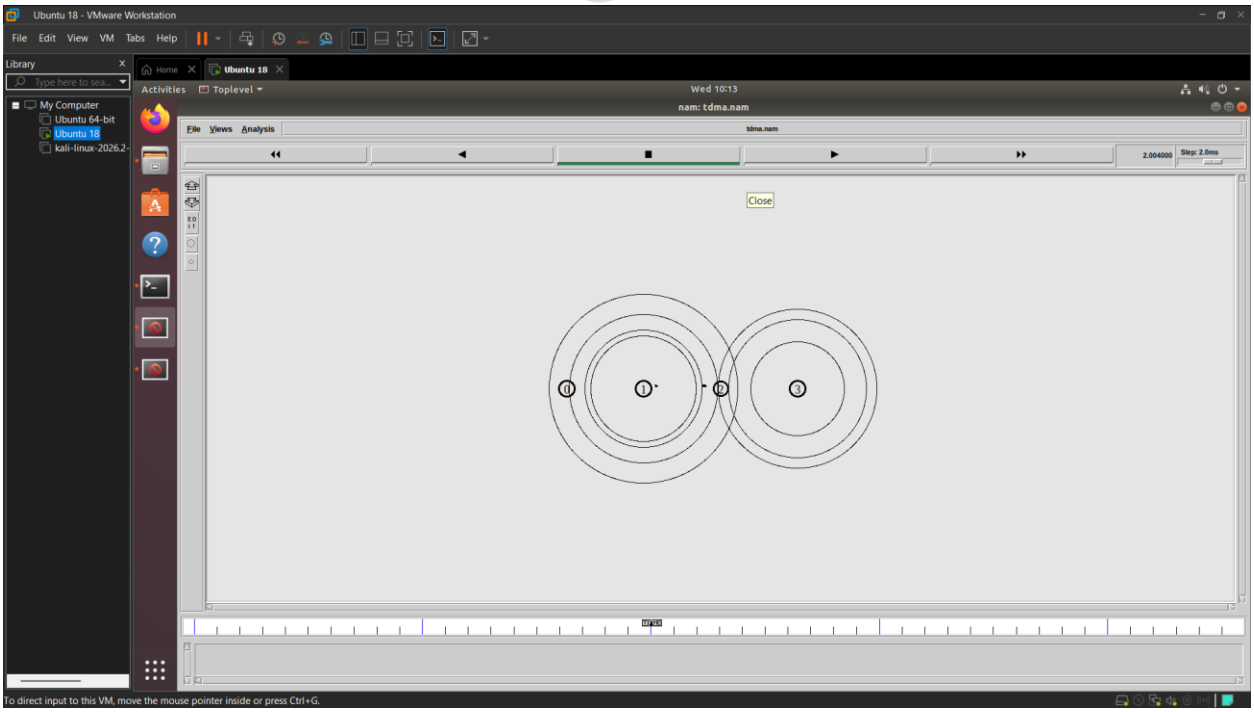
# Slot 2 : Node1 transmits
nsat2.0"cbr1 start"
nsat2.5"cbr1 stop"

# Slot 3 : Node2 transmits
nsat3.0"cbr2 start"
nsat3.5"cbr2 stop"
# Slot 4 : Node0 transmits again
```

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```
nsat4.0"cbr0 start"
nsat4.5"cbr0 stop"
# ================================================
# FINISH
# ================================================
proc finish {} {
    global ns tracefd namfd
    $ns flush-trace
    close $tracefd
    close $namfd
    puts "Simulation Completed"
    exec nam tdma.nam &
    exit 0
}
$ns at 5.0 "finish"
puts "Starting TDMA Simulation..."
$ns run
```

OUTPUT:



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Observation

- Four wireless nodes were created.
- Node 3 acted as the destination node.
- Nodes 0, 1, and 2 transmitted data in separate time slots.
- No simultaneous transmissions occurred.
- Packet collisions were avoided due to slot allocation.
- Data packets successfully reached the destination node.
- NAM animation clearly showed sequential transmissions.

Result

The TDMA-style MAC protocol simulation was successfully implemented. Scheduled transmissions occurred according to assigned time slots, and packets were delivered successfully to the destination node.

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